

1.1. Part 1: Objective

Before selecting a controller, the first question is whether the plant, sensor timing, and disturbance spectrum permit the required disturbance rejection at all. The coarse gimbal is torque-driven, the camera samples line-of-sight (LOS) at 60 Hz with significant latency, and the platform disturbance contains broadband energy together with drifting tones near 22 Hz and 47 Hz.

The purpose of Part 1 is therefore to determine the disturbance-rejection bandwidth that is physically achievable with camera-only feedback, and to locate the important disturbance components relative to that bandwidth.

1.2. Plant model

The single-axis torque-to-gimbal-angle plant is represented by a rigid-body double integrator, two lightly damped structural modes, actuator current-loop lag, and transport delay:

$$P(s) = [1/(J s^2)] * \prod_k [\omega_k^2 / (s^2 + 2 \zeta_k \omega_k s + \omega_k^2)] * [1/(1 + \tau s)] * e^{-T_d s}$$

The parameters used in the analysis are $J = 2.5 \times 10^{-3} \text{ kg m}^2$, structural modes at 75 Hz ($\zeta = 0.03$) and 220 Hz ($\zeta = 0.04$), actuator time constant $\tau = 0.2 \text{ ms}$, and actuator transport delay $T_d = 0.5 \text{ ms}$.

1.3. Rigid-body behaviour

At frequencies well below the first structural mode, the flexible-mode terms are approximately unity and the actuator lag is small. The plant therefore reduces to the rigid-body approximation:

$$P_{rb}(s) \sim 1/(J s^2) = 400/s^2$$

This is a double integrator. Its magnitude falls at 40 dB/decade and its phase is approximately -180 degrees. This is important because the rigid body already consumes most of the phase available before instability; controller phase lead and delay management therefore become central to the crossover design.

1.4. Structural modes

The plant contains two lightly damped modes. The first is at 75 Hz with damping ratio 0.03, giving an approximate quality factor $Q = 1/(2\zeta) = 16.7$. The second is at 220 Hz with damping ratio 0.04, giving $Q = 12.5$.

The 75 Hz mode is the more important bandwidth constraint because it lies close to the disturbance band and is strongly resonant. Carrying substantial loop gain toward this mode makes the closed loop sensitive to modal-frequency and damping uncertainty and can introduce additional unity-gain crossings.

1.5. Actuator lag and delay

The actuator current-loop time constant is 0.2 ms, corresponding to an approximate bandwidth of $1/(2\pi\tau) = 796 \text{ Hz}$. The current loop is therefore much faster than the expected

coarse-loop crossover and is not the primary bandwidth limitation. The actuator also contributes 0.5 ms of transport delay.

1.6. Camera timing and observability

The camera runs at 60 Hz, so its Nyquist frequency is 30 Hz. Camera-only feedback therefore cannot uniquely observe physical content above 30 Hz. In particular, the 47 Hz tone aliases to 13 Hz:

$$f_{alias} = |47 - 60| = 13 \text{ Hz}$$

A measured 13 Hz component cannot be distinguished from a genuine 13 Hz disturbance or an aliased 47 Hz disturbance without additional information. This is an information limitation rather than a controller-tuning problem.

1.7. Camera delay budget

A 60 Hz sampled camera also incurs sample-and-hold age. Using half a frame period gives $T_{ZOH} = 1/(2 \times 60) = 8.33 \text{ ms}$. With the conservative 90th-percentile camera latency of approximately 35.63 ms, the effective camera timing budget is:

$$T_{eff} \sim 8.33 \text{ ms} + 35.63 \text{ ms} = 43.96 \text{ ms}$$

This large effective delay is the dominant reason the camera-only loop must remain low bandwidth.

1.8. Delay-limited crossover

A pure delay contributes phase $\phi_d = -\omega T$. If approximately 45 degrees of phase budget is reserved for delay, then $\omega_c T$ should remain below roughly $\pi/4$. With $T = 43.96 \text{ ms}$:

$$\omega_c \leq (\pi/4)/T \sim 17.87 \text{ rad/s}$$

$$f_c \leq \omega_c/(2\pi) \sim 2.84 \text{ Hz}$$

A realistic robust camera-only crossover is therefore on the order of 2-3 Hz. Less conservative latency assumptions may move the value somewhat, but not close to the 22 Hz or 47 Hz tones.

1.9. Sensitivity and achievable rejection bandwidth

For loop transfer $L(s) = P(s)C(s)$, the output-disturbance sensitivity is:

$$S(s) = 1 / [1 + L(s)]$$

Disturbance rejection requires $|S(j\omega)| < 1$, or below 0 dB. For the classical camera-only design, the measured crossover is approximately 2.81 Hz and the peak sensitivity is $M_s = 3.81$ (about +11.6 dB). The -3 dB rejection bandwidth is only about 1.27 Hz.

At the important tones, $|S(22 \text{ Hz})|$ is approximately 1.156 (+1.26 dB), so the 22 Hz component is slightly amplified rather than rejected. At 47 Hz, $|S(47 \text{ Hz})|$ is approximately 1.003 (+0.03 dB), so the 47 Hz component passes essentially unchanged.

1.10. Fundamental disturbance-rejection limit & conclusion

The 60 Hz camera-only coarse-gimbal loop is fundamentally limited to low-frequency disturbance rejection, with useful rejection up to approximately 1.27 Hz and a practical crossover near 2.81 Hz. Consequently, the 22 Hz tone lies far outside the controllable bandwidth, while the 47 Hz tone is additionally above the camera's 30 Hz Nyquist frequency and aliases to 13 Hz, preventing unique reconstruction. Significant broadband disturbance energy also extends to approximately 80 Hz, well beyond the achievable camera-only bandwidth.

Increasing crossover is not a viable solution because sensitivity amplification moves into the disturbance-rich frequency range and increasing loop gain approaches the lightly damped 75 Hz structural mode, reducing robustness. The fundamental limitation is therefore camera sampling and latency combined with the structural dynamics, rather than actuator authority.

Thus, camera-only feedback should not be expected to achieve meaningful rejection of the 22 Hz and 47 Hz disturbances. This establishes the design direction for the subsequent parts: Part 2 demonstrates the camera-only control ceiling, while Part 3 introduces faster inertial sensing to make high-frequency disturbance information observable and usable for feedback.

2.1. Part 2: Objective

Part 2 implements a single-stage discrete-time feedback controller for the coarse gimbal using the camera as the LOS measurement. The target is approximately 50 microrad RMS, starting from the much larger open-loop vibration level, while respecting the 60 Hz camera timing, drifting 22 Hz and 47 Hz tones, lightly damped structural modes, actuator limits, and friction/cogging.

The delivered controller is a classical two-stage lead compensator with a 20 Hz high-frequency roll-off pole. The objective of the implementation is to determine how far a robust camera-only loop can actually go.

2.2. Controller architecture & design

The Part 2 controller is a camera-only, single-stage discrete-time feedback loop. The LOS reference is zero, and the 60 Hz camera provides the feedback measurement including its sampling, latency, and noise. The resulting error is processed by the controller to generate a torque command, which is limited before passing through the actuator and gimbal dynamics. Platform vibration enters as an output disturbance, while friction and cogging act through the mechanical torque path. Only simulated sensor measurements are available to the controller; simulation truth is not used. The controller is a two-stage lead compensator with high-frequency roll-off, designed in continuous time and subsequently discretized:

$$C(s) = K_p [C_{lead}(s)]^2 C_{rolloff}(s) \quad , \text{ where,}$$

$$K_p = 0.1724$$

$$C_{lead}(s) = (1 + 0.110883 s) / (1 + 0.0138604 s)$$

$$C_{rolloff}(s) = 1 / (1 + s/125.66) \quad , \text{ therefore,}$$

$$C(s) = 0.1724 [(1 + 0.110883 s)/(1 + 0.0138604 s)]^2 [1 / (1 + s/125.66)]$$

There is no separate lag term ($C_{lag} = 1$), so the implemented controller is essentially a two-stage lead controller with roll-off.

Two lead stages are required because, below the first structural mode, the gimbal behaves approximately as a rigid-body double integrator, $P_{rb}(s) \sim 400/s^2$ which already contributes approximately -180° of phase. Camera timing, actuator lag, and transport delay introduce additional phase lag. For $\alpha = 8$, each lead stage can provide up to approximately 51.1° of phase lead. Using two stages distributes the required phase compensation without requiring an excessively aggressive single stage. Each stage has a zero near 1.44 Hz, a pole near 11.48 Hz, and maximum phase lead near 4.07 Hz, placing the phase support around the intended low-frequency crossover.

Finally, a 20 Hz roll-off pole ($\omega_r = 125.66 \text{ rad/s}$) attenuates controller gain above the useful control band. This prevents the lead stages from carrying excessive gain toward the lightly damped 75 Hz structural mode, balancing phase recovery near crossover with high-frequency robustness.

2.3. Why conservative design?

Increasing crossover was evaluated directly. Performance deteriorated as loop gain approached the lightly damped 75 Hz mode:

Target crossover	L at 75 Hz	LOS RMS	Unity crossings	Peak torque
2 Hz	0.07	236.0 microrad	1	2.2 mN m
5 Hz	0.74	254.2 microrad	1	13.8 mN m
10 Hz	1.98	279.6 microrad	3	41.8 mN m
20 Hz	5.61	636.0 microrad	3	260 mN m

At 2 Hz the structural mode remains well outside significant loop gain. By 10 Hz the loop has gain greater than unity around the 75 Hz mode and develops multiple unity-gain crossings. The low-bandwidth design is therefore a stability and robustness choice, not merely conservative gain tuning.

2.4. Closed-loop sensitivity, tonal rejection & actuator limits

For loop transfer $L(s) = P(s)C(s)$, rejection of output disturbances is governed by the sensitivity function, $S(s) = 1/[1 + L(s)]$, where useful disturbance rejection requires $|S(j\omega)| < 1$. The implemented camera-only controller has a crossover near 2.81 Hz, a -3 dB rejection bandwidth of approximately 1.27 Hz, and peak sensitivity $M_s = 3.81$ (+11.6 dB). Consequently, effective rejection is confined to very low frequencies.

This bandwidth limitation is particularly important for the drifting 22 Hz and 47 Hz tonal disturbances. A fixed notch was deliberately avoided because a narrow notch would lose alignment as the tones drift, while a wider notch would affect a larger frequency range and consume phase margin. The 47 Hz tone also presents a fundamental camera-only limitation: with a 60 Hz sampling rate, the camera Nyquist frequency is 30 Hz and the tone aliases to, $f_{alias} = 13 \text{ Hz}$. The camera therefore cannot uniquely distinguish a genuine 13 Hz component from the aliased 47 Hz disturbance. The closed-loop sensitivity confirms the resulting lack of tonal rejection:

$$|S(22 \text{ Hz})| \approx 1.156 = +1.26 \text{ dB}, |S(47 \text{ Hz})| \approx 1.003 = +0.03 \text{ dB}$$

Thus, the 22 Hz tone is slightly amplified and the 47 Hz tone passes essentially unchanged, consistent with both tones lying well above the useful rejection bandwidth.

Actuator limits are explicitly included in the nonlinear simulation: torque is saturated at $\pm 0.5 \text{ N}\cdot\text{m}$ and gimbal angular rate at $\pm 3 \text{ rad/s}$, $\tau_{sat} = \text{clip}(\tau_{cmd}, -0.5, +0.5)$.

However, neither limit is active in the nominal Part 2 run. Peak commanded torque is only approximately 2.2 mN·m, about 0.44% of the torque limit, with 0% torque saturation and 0% rate-limit activity. Furthermore, the controller contains no free pure-integrator term, so classical integrator wind-up is not a significant concern. Therefore, actuator saturation and wind-up do not explain the performance shortfall; the dominant limitation is the camera-only rejection bandwidth and inability to adequately observe and reject the high-frequency tonal disturbances.

2.5. Part 2: Conclusion

The two-stage lead controller with high-frequency roll-off does not achieve the 50 μrad RMS target, with the nonlinear simulation remaining at approximately 236 μrad RMS. The primary limitation is the camera-only feedback architecture: camera delay restricts robust crossover to approximately 2.8 Hz, placing the 22 Hz tone well outside the rejection bandwidth, while the 47 Hz tone exceeds the 30 Hz camera Nyquist frequency and aliases to 13 Hz. Increasing crossover does not solve the problem because it raises loop gain toward the lightly damped 75 Hz structural mode, causing sensitivity amplification and multiple unity-gain crossings.

The controller therefore provides a simple and transparent demonstration of the camera-only performance ceiling predicted in Part 1. Its inability to reject the dominant tones motivates Part 3, where a faster inertial sensor is introduced to provide timely vibration-band information.

3.1. Part3: Objective

Part 2 established that the camera-only controller is fundamentally limited by the information available to the feedback loop. The 60 Hz camera provides an absolute line-of-sight (LOS) measurement, but its sampling rate and latency restrict useful feedback bandwidth to only a few hertz. The 22

Hz tonal disturbance lies well outside that useful rejection region, while the 47 Hz tone lies above the camera Nyquist frequency.

Part 3 therefore addresses the sensing and timing problem rather than simply increasing controller gain. The controller remains the two-stage lead controller with high-frequency roll-off developed in Part 2. The principal change is the measurement supplied to that controller: camera and gyro information are fused into a timing-aware current-time LOS estimate.

Key distinction: latency limits how quickly camera information can be used; sampling limits what physical information the camera contains.

3.2. Sensor selection & roles

The three available sensors provide different information about the LOS dynamics:

Sensor	Primary strength	Primary limitation	Selected role
Camera	Absolute LOS	60 Hz, large/jittered latency	Low-frequency absolute reference
Gyro	1 kHz inertial rate	Bias/drift and finite latency	Fast vibration-band information
Encoder	1 kHz gimbal angle	Does not directly measure complete inertial LOS	Plant-state/validation support

The camera and gyro are selected as the primary fusion pair because their measurements are complementary. The camera directly measures the controlled quantity—absolute LOS—and provides the low-frequency reference needed to prevent long-term drift and correct accumulated gyro bias. However, its low sampling rate and latency make it unsuitable for observing fast vibration.

The gyro provides the missing high-frequency information. At approximately 1 kHz sampling, its Nyquist frequency is about 500 Hz, so both the 22 Hz and 47 Hz tones are directly observable. In particular, it measures the physical 47 Hz motion without the camera's 13 Hz alias ambiguity and provides lower-age measurements between camera frames. However, because the gyro measures angular rate rather than absolute angle, its integrated estimate is susceptible to bias and long-term drift. It therefore complements rather than replaces the camera.

The encoder provides accurate mechanical gimbal angle and is useful for plant-state reconstruction, model validation, and actuator monitoring. However, encoder angle alone does not represent the complete inertial LOS. It is therefore retained as an auxiliary sensor rather than the primary fusion measurement.

Overall, the sensor responsibilities are, camera for low-frequency LOS and gyro for fast platform vibration information.

This makes camera + gyro fusion the most direct architecture for overcoming the camera timing and sampling limitations identified in Part 2.

3.3. Fusion architecture & Kalman estimator

The Part 3 architecture changes only the measurement path; the two-stage lead controller from Part 2 is retained. Instead of acting directly on the delayed 60 Hz camera measurement, the controller receives a current-time LOS estimate obtained by fusing the camera and 1 kHz gyro measurements. The key architectural transformation is therefore, $\theta_{camera, delayed} \rightarrow \hat{\theta}_{LOS, current}$.

A Kalman estimator is used because the camera and gyro operate at different rates, have different noise characteristics, and provide complementary state information. The estimator propagates at the fast control rate. Frequent gyro updates constrain the high-frequency motion between camera frames, while camera updates provide the absolute low-frequency LOS correction and prevent unbounded drift caused by gyro bias. The resulting fused state is propagated to the current control time before being supplied to the lead controller, allowing the feedback loop to operate on a more timely LOS estimate rather than a stale camera sample.

3.4. Timing validation and prediction

Tone-by-tone validation revealed that the Kalman estimate was referenced to the delayed measurement timestamp rather than the current actuation time, producing approximately 2 ms of effective lag:

Frequency	Lag before	Lag after	Amplitude error after
22 Hz	1.41 ms	0.04 ms	-29%
47 Hz	2.00 ms	0.80 ms	-36%
75 Hz	2.14 ms	1.54 ms	-38%

The nearly constant initial lag identified a measurement/transport-age problem rather than normal filter roll-off. The estimator was therefore propagated forward by two control steps to obtain an estimate at the actuation instant.

This almost eliminated the timing error at 22 Hz, reducing the lag from 1.41 ms to 0.04 ms. At higher frequencies, some lag and amplitude attenuation remained because stochastic vibration cannot be predicted perfectly. Thus, prediction introduces a fundamental trade-off between lower effective latency and increasing prediction uncertainty.

Independent validation confirmed the benefit of fusion: under noiseless conditions, the fused estimator achieved approximately 27 μ rad error for a ~233 μ rad signal, compared with approximately 382 μ rad for camera-only estimation. This validates the fusion approach while also showing that improved state estimation does not by itself guarantee equivalent closed-loop LOS performance.

3.5. Part 3: Conclusion

Camera-gyro fusion successfully addresses the camera-only sensing limitation: the camera provides absolute LOS while the

gyro supplies fast inertial information. The timing-aware Kalman estimator further reduces measurement age, decreasing the 22 Hz equivalent lag from 1.41 ms to 0.04 ms.

However, the 15 μ rad RMS target is not achieved because, once sensing limitations are reduced, performance becomes constrained by broadband disturbance predictability, structural dynamics, finite mechanical bandwidth, and nonlinear plant effects. Thus, fusion shifts the system from a sensing-limited regime to a plant/prediction-limited regime.

4.1. Part 4: Dual stage architecture

The dual-stage architecture separates large-range pointing from high-precision stabilization as shown in Figure 1. The camera and gyro are fused by the timing-aware Kalman estimator to drive the coarse gimbal, which handles acquisition with a bandwidth of approximately 2–3 Hz. The coarse stage only needs to bring the beam close enough for the fine stage to take over; it is not expected to achieve the final 15 μ rad RMS requirement by itself.

The FSM/QPD stage provides the high-bandwidth correction. Once the coarse gimbal has positioned the beam near the QPD, the fused LOS estimate is mapped into an FSM pre-position command during fine capture. When the beam is safely within the QPD's ± 200 μ rad linear range, preferably within desired safety margin, control is smoothly handed over to the QPD loop. The QPD measures the residual beam-position error, the fine controller generates the FSM command, and the FSM rapidly corrects vibration and jitter with a target bandwidth of approximately 300 Hz.

Because the FSM has limited angular range, its slow mean displacement is passed through a 0.3–0.5 Hz low-pass offload path and added to the coarse-gimbal command. The gimbal gradually takes over this slow correction, allowing the FSM to remain near its center and preserve range for high-frequency disturbances. Anti-windup and FSM-range supervision prevent the fine controller from remaining saturated during large disturbances or acquisition transients.

The overall division of responsibility is therefore:

Gimbal = wide range, acquisition and slow correction
 FSM/QPD = small range, high-speed precision correction

This bandwidth separation prevents the two stages from competing with each other while avoiding the 75 Hz coarse-gimbal structural limitation identified earlier. The offload loop prevents FSM saturation, while the staged acquisition and handoff logic avoids engaging the QPD close to its nonlinear operating boundary.

5.1. Part 5: Validation methodology

For validation, I would build two complementary test subsystems.

The first would reproduce the mechanical disturbance environment. A two-axis motion stage would excite the terminal

with representative platform motion, while an electromechanical shaker would be used to excite the gimbal structure around its dominant resonant frequencies. A laser vibrometer would independently measure the resulting platform and structural vibrations. The measured vibration spectrum would then be compared with the commanded disturbance profile to verify that the bench accurately reproduces the intended real-world environment.

The second subsystem would provide a controlled optical target and independent ground truth. A steerable laser source would generate the incident beam at known pointing angles, while an independent laser measurement system in the optical path would measure the actual beam angle and serve as the reference measurement.

The terminal would then run the complete PAT controller to acquire and track the beam using the gimbal and FSM. Pointing performance would be calculated from the difference between the independently measured beam angle and the terminal's achieved pointing angle. In addition to RMS error, I would evaluate the mean error, standard deviation, and error spectrum across progressively more challenging conditions: quiet platform → tonal disturbances → broadband vibration → structural-mode excitation. I would also repeat the tests over a range of incident beam angles to verify that tracking performance is maintained across the terminal's operating field of regard.

If a test fails, I would first focus on root-cause identification rather than immediately retuning the controller. During each run, I would record the optical LOS reference, camera/gyro/QPD measurements, gimbal and FSM commands and positions, accelerometer data, saturation and offload states, and controller timing. Comparing these signals in the time and frequency domains, especially using residual LOS PSD and coherence, would help isolate the source of the error.

For example, a peak appearing in both LOS error and structural accelerometer measurements near 75 Hz would indicate a mechanical resonance or insufficient isolation. Increased residual error around loop crossover, together with phase loss or excessive actuator demand, would indicate a control limitation. If the independent optical measurement is accurate but the camera or QPD estimate is noisy, biased, or delayed, the problem would more likely lie in sensor calibration, timing, or estimation.

Once the terminal can reliably acquire targets across different pointing angles, hand over to fine tracking, and maintain lock under representative vibration and structural excitation, I would consider the control architecture technically validated. I would then move to field testing under more realistic operating conditions, including strong winds, reduced visibility, rain or storms, and varying ambient-light conditions. The system would be considered ready to ship once it consistently meets the required pointing and tracking performance across these mechanical, optical, and environmental conditions.

6.1. Part 6: Reflection

The main limitation is that the coarse stage still falls well short of the required pointing accuracy. Camera-only control remains around 236 μrad RMS, and although camera-gyro fusion improves state estimation and reduces timing error, it does not bring the coarse gimbal close to the 15 μrad RMS target. The remaining error is dominated by structural dynamics, broadband vibration, prediction uncertainty, and nonlinear effects such as friction and cogging.

With another week, I would focus on understanding and reducing the predictable part of that residual error, rather than pushing the feedback bandwidth higher. In particular, I would investigate predictive/feedforward cancellation of the 22 Hz and 47 Hz disturbances and position-dependent cogging compensation. The goal would be to quantify how much of the LOS error can be anticipated and cancelled proactively, leaving the feedback loop to handle only the unpredictable residual.

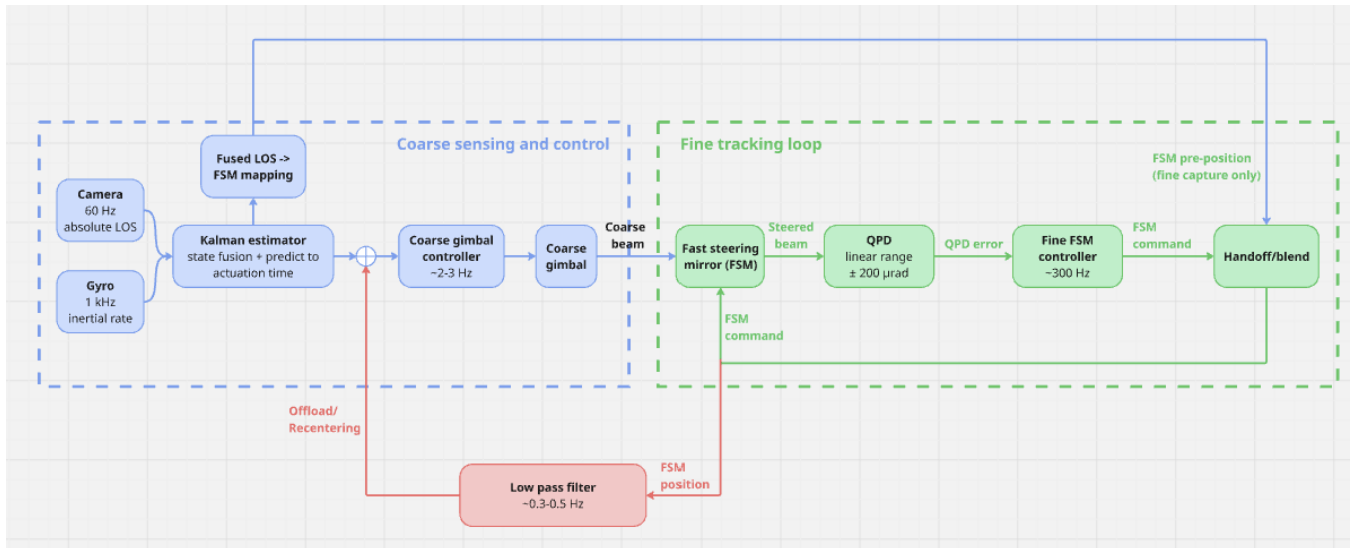


Figure 1: Dual stage architecture

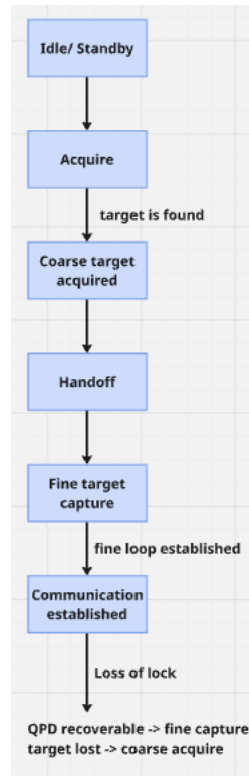


Figure 2: Software state machine